

THE CONTROL DECIDES THE ANSWER: A MATCHED-OPERATOR NULL FOR EXPLANATION CHANGE UNDER MODEL UPDATES

Anonymous authors

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ABSTRACT

When a deployed model is updated in response to a distribution shift, its feature attributions move. Existing work measures how far they move and treats the residual, after conditioning on prediction change, as instability. That is not yet a criterion, because it carries no account of how far an explanation *should* have moved. We supply one. Using shift families whose Bayes-optimal predictor is available in closed form, we define a warranted-change reference ω and decompose attribution change across an update into warranted, unwarranted and explainer-noise parts. The design turns on a control prior work does not use: a **matched-operator null**, applying the identical update to fresh *source* data, so treatment and control differ in the distribution alone and not in the amount of training. This changes the conclusion. Across 681 runs the null reproduces a median 82% of the movement that follows an update. On a no-shift placebo, where the answer is fixed by construction, it returns 1.02 on an MLP and 0.98 on gradient-boosted trees with exact TreeSHAP, while the seed-retraining floor prior work uses implicitly returns 0.32 and 1.90 on the same data, disagreeing with itself sixfold. We state the consequence for our own results instead of leaving it to be found: against that seed floor our headline effect is inverted (median 0.31), so the case for the matched null rests on the placebo. Applying the missing control to a published audit, 64% of its 45 settings produce movement smaller than refitting on a resample of the same data. Unwarranted change is real but bounded: present across MLPs over a $630\times$ parameter range, trees and a small ResNet, weaker on real census covariates, absent on DistilBERT. Magnitude indicates legitimacy only once an update completes the adaptation it implies, so it fails in the light-update regime auditors work in.

1 INTRODUCTION

Post-hoc feature attributions are consumed as evidence: a practitioner reads which features a model relied on and acts on what they read. Models do not stand still. They are fine-tuned when data drifts, retrained on refreshed samples and redeployed. Whether the evidence survives such an update is an open question.

The question has been asked in pieces. Mougán et al. (2023) compare explanation *distributions* across domains for a frozen model. Subramaniakuppasamy & Gajjar (2026) filter to prediction-preserved pairs before measuring attribution stability under input perturbation. Hemmat & Fatemi (2025) difference attributions across a model update. Dhayalkar (2026) tracks attribution change across fine-tuning epochs. Each measures a magnitude. None supplies the prior quantity, which is how far the explanation should have moved. Absent that quantity, every method treats change as damage and names the residual by fiat, calling it instability, semantic drift or risky reliance redistribution.

Treating all change as damage misclassifies a case that occurs in practice. A model fine-tuned on data from which a spurious shortcut has been removed should stop attributing to that shortcut. Its predictions on most inputs are unchanged, its explanation changes a great deal, and the change is the

desired behaviour. A metric that scores this as instability is measuring something other than what it claims.

Two checkpoints that agree on the shared support but differ in their gradients are different functions that the data did not distinguish, which is the underspecification phenomenon (D’Amour et al., 2022) read through attributions. There is a second gap, and it proves to be the larger one. Measuring how far attributions moved after an update requires knowing how far they would have moved anyway. Prior work’s implicit control is an independent retrain with a different seed. That control differs from the treatment in operator, data volume and step count at once, so a comparison against it does not isolate the distribution. We supply a control that differs in the distribution alone, and find that the choice reverses the ordering of update regimes.

Contributions.

1. **A warranted-change reference.** We define ω as the change in the attributions of the *Bayes-optimal predictor* of each environment, not of the causal generating mechanism. On generators where that predictor is available in closed form, ω is exact: identically zero under covariate shift, non-zero and computable under concept shift and shortcut removal. The definitional choice is what makes the shortcut case come out right (§3).
2. **A matched-operator null.** The control for a fine-tune-on-target treatment is the identical fine-tune on fresh *source* data, matched in learning rate, epochs, step count and sample size (§3, §6.2).
3. **A decomposition, not a metric.** Every distance we use already exists (Alvarez-Melis & Jaakkola, 2018; Yeh et al., 2019; Bhatt et al., 2020; Agarwal et al., 2022a;b; Hedström et al., 2023); the contribution is the reference and the controls.
4. **Three propositions** on which explainer classes inherit stability from prediction stability, each with a sharpness construction and a per-point numerical check (§4).
5. **An empirical finding that survives its own controls**, together with an explicit account of where it stops holding (§6).

What we do not claim. We do not propose a new stability metric. We do not detect distribution shift; that is the contribution of Mougán et al. (2023), whose frozen-model setting is not ours. We were not first to condition on prediction preservation; Subramaniakuppusamy & Gajjar (2026) were, and we adopt their conditioning. We avoid the phrase “explanation drift”, which denotes at least three distinct objects in the recent literature (Appendix A).

2 SETUP AND RELATED WORK

A category error worth stating once. For a frozen model f and a deterministic explainer E , the explanation $E_f(x)$ is a fixed function of x . Changing $P(X)$ does not change $E_f(x)$ at any x ; it changes only which x are observed. Instance-level explanation change under pure data shift with a frozen model is identically zero, and the only measurable object is the shift of the explanation *distribution*, which is what Mougán et al. (2023) measure. Genuine instance-level change requires one of: the model changed, the input changed, or the explainer is stochastic. This paper studies the first. The second belongs to Subramaniakuppusamy & Gajjar (2026) and the adversarial-fragility line (Ghorbani et al., 2019; Dombrowski et al., 2019; Slack et al., 2020). The third is a nuisance we measure and subtract. We verify all three empirically instead of asserting them (§6.1).

Objects. $f : \mathcal{X} \rightarrow \mathbb{R}$ is the explained scalar output, always the logit, so that the theory, the explainers and the closed-form reference share a space. An update operator produces $f_{t+1} = \mathcal{U}(f_t, \mathcal{D})$. A probe set $\mathcal{P}$ is drawn from the *shared support* of source and target, so neither checkpoint extrapolates on it, and the prediction-preserved subset is $\mathcal{P}_\epsilon = \{x \in \mathcal{P} : |p_t(x) - p_{t+1}(x)| \leq \epsilon\}$. Conditioning on preservation is not optional: Shapley efficiency implies that if two inputs’ predictions differ their attribution vectors must differ in at least one coordinate, so measuring change without conditioning partly measures an algebraic necessity. Shifts are *generated*, never assumed. Covariate shift moves $P(X)$ with $P(Y | X)$ fixed, concept shift moves $P(Y | X)$, and shortcut removal severs a feature that was predictive but never causal.

Table 1: Related work organised by what changes. The last row is closest to this paper.

What changes	Work	What is missing
data, model frozen the input	Mougan et al. (2023) Subramaniakuppusamy & Gajjar (2026); Ghorbani et al. (2019)	instance-level change is vacuous here single model; residual named by fiat
the seed the model class	Bensmail (2025); Laberge et al. (2023) Thackshanaramana B. (2026)	no shift; this is our <i>floor</i> architecture varies, distribution does not
the checkpoint	Hemmat & Fatemi (2025); Dhayalkar (2026); Elangovan & Ting (2026)	no reference, no matched control

A separate line explains the shift itself instead of the model’s response to it (Kulinski & Inouye, 2023; Hinder et al., 2023), and a further line documents that explainers disagree with one another on a fixed model (Krishna et al., 2022). Neither supplies a reference for how far an explanation should move. We exclude attention weights, whose status as explanation is contested (Jain & Wallace, 2019). Table 1 organises the field by the object that changes. The two nearest works deserve specifics. Hemmat & Fatemi (2025) difference occlusion attributions across model pairs and label movement uncorrelated with behaviour change “spurious”, which is a heuristic and not a reference, and which misfires on the shortcut case. Subramaniakuppusamy & Gajjar (2026) establish that without conditioning on prediction preservation up to 99% of compared pairs have changed predictions, then measure raw distance on what remains, with no floor for comparison.

3 WARRANTED CHANGE AND THE TWO CONTROLS

The reference. We define ω against the Bayes-optimal predictor $h_e(x) = \text{logit } P_e(Y=1 \mid X=x)$ of each environment:

$$\omega(x) = d(\phi(\text{IG}[h_S](x, b)), \phi(\text{IG}[h_T](x, b))). \quad (1)$$

Referencing the *causal* mechanism instead would invert the shortcut verdict. Under shortcut removal the causal function is unchanged, yet a correct model must stop using the shortcut, so a causal reference would label that change unwarranted. The Bayes reference is exactly zero under covariate shift and available in closed form under concept shift and shortcut removal.

The floors. ν is the explainer’s own noise, measured between two runs on one checkpoint. ρ_{null} is the matched-operator null: the identical update applied to fresh source data, holding learning rate, epochs, step count and sample size fixed, so that only the sampling distribution differs. ρ_{seed} is the independent-retrain floor that prior work uses. We report

$$\text{UEC} = \mathbb{E}_{\mathcal{P}_e}[\Delta] - \omega - \max(\nu, \rho_{\text{null}}), \quad \text{ratio} = \mathbb{E}[\Delta] / \rho_{\text{null}}. \quad (2)$$

UEC is a difference and not a ratio to output change, because such a ratio (Agarwal et al., 2022a) is undefined and explosive in the prediction-preserving regime this paper concerns. The null and the treatment contain the same optimisation transient, and only the treatment additionally contains the distribution signal. Comparing a fine-tune against a from-scratch retrain, as the seed floor does, confounds the operator with the distribution.

4 THEORY

Let $\delta = f_t - f_{t+1}$, baseline b , path $\gamma(\alpha) = b + \alpha(x - b)$. All explainers below are linear in the explained function.

Proposition 1 (Path-integrated: the aggregate is inherited, the allocation is not). (i) $\sum_j \Delta \text{IG}_j(x) = \delta(x) - \delta(b)$ exactly, hence $|\sum_j \Delta \text{IG}_j(x)| \leq 2\epsilon$ when the checkpoints agree to ϵ at x and at b , requiring agreement at the two endpoints only and not along the path. (ii) $\|\Delta \text{IG}(x)\|_1 \leq \|x - b\|_1 \sup_{\gamma} \|\nabla \delta\|_{\infty}$. (iii) Sharpness: output preservation alone does not bound individual components. With $\delta(z) = \epsilon \sin(\omega(z_1 - z_2))$, $b = 0$, $x = \mathbf{1}$, we have $|\delta| \leq \epsilon$ everywhere, yet $\Delta \text{IG}_1 = \epsilon \omega$ is unbounded while the aggregate is zero.

Proposition 2 (Local gradients inherit nothing, not even the aggregate). *With $\delta(z) = \epsilon \sin(\omega z_1)$, the aggregate gap $\sum_j \Delta(G \times I)_j(x) = x_1 \epsilon \omega$ is unbounded under the premise that pins IG’s aggregate to 2ϵ .*

Proposition 3 (Shapley, under a coalition-level premise). *If $|v_f(S) - v_{f'}(S)| \leq \epsilon_{\text{coal}}$ for every coalition S , then $\|\varphi^f - \varphi^{f'}\|_\infty \leq 2\epsilon_{\text{coal}}$.*

Remark. ϵ_{coal} is a supremum over masked, off-manifold inputs. Fine-tuning optimises a loss over the data distribution and leaves the model unconstrained there, so preservation on the data implies nothing about ϵ_{coal} . We measure the gap rather than assume it (§6.6).

Corollary. Under a prediction-preserving update, IG’s aggregate attribution mass is bounded, gradient $\times$ input’s is not, and Shapley’s is bounded only if the coalition premise holds. Collinearity constrains the allocation further, since no ranking can be simultaneously faithful, stable and complete when features are collinear (Caraker et al., 2026); our redundant-feature block is built to exercise that case. No class is protected in its *allocation* across features, which is the part a practitioner reads. The sharpness construction is not exotic in kind: it is a difference that oscillates transverse to the integration path while vanishing on it, which is to say two checkpoints that agree on the data manifold and curve differently off it. This is the mechanism Dombrowski et al. (2019) identified in input space, transposed to model space.

5 EXPERIMENTAL PROTOCOL

Shifts are generated from a structural equation model with $d=20$ features in causal, shortcut, redundant and noise blocks, for which the Bayes-optimal logit and its integrated gradients have closed forms. Attributions target the logit; the attribution baseline is fixed across checkpoints, since a per-checkpoint baseline makes Δ IG uninterpretable. Activations are smooth by default, because on a ReLU network IG completeness holds only to $O(1/n_{\text{steps}})$ and would contaminate the Proposition 1 check. Gradient-boosted trees use XGBoost with exact TreeSHAP (Chen & Guestrin, 2016). We use seven explainers spanning path-integrated, local-gradient and Shapley-type classes (Sundararajan et al., 2017; Erion et al., 2021; Simonyan et al., 2013; Smilkov et al., 2017; Lundberg & Lee, 2017; Ribeiro et al., 2016; Lundberg et al., 2020), implemented through Captum (Kokhlikyan et al., 2020). Beyond the generator we use ACS Income (Ding et al., 2021), CIFAR-10 with common corruptions (Krizhevsky, 2009; Hendrycks & Dietterich, 2019), and DistilBERT on IMDB (Sanh et al., 2019; Maas et al., 2011). Tests are paired across seeds with Holm correction (Holm, 1979) and Cliff’s δ (Cliff, 1993). Our reading of what an attribution evaluation must control for follows recent critiques of the evaluation protocols themselves (Kiourti et al., 2025; Sridhar et al., 2026). Figures are drawn to be legible under the common colour-vision deficiencies and in monochrome (Appendix C).

6 RESULTS

6.1 THE CONTROLS BEHAVE, AND THE PLACEBO IS DECISIVE

Deterministic explainers reproduce bitwise on a fixed checkpoint, so $\nu = 0$ exactly; a frozen model under pure data shift yields identically zero instance-level change, confirming §2 numerically and not by assertion. On a no-shift placebo, where the correct answer is fixed in advance, the matched null returns 1.02 on an MLP and 0.98 on gradient-boosted trees with exact TreeSHAP. The seed floor returns 0.32 and 1.90 on the same data. It disagrees with itself sixfold across two model classes and errs in opposite directions, so no fixed correction recovers it. This test does not depend on any shift result and was specified before them.

6.2 WHICH CONTROL YOU CHOOSE REVERSES THE CONCLUSION

Over a grid of shift magnitude $\times$ update strength (600 runs, 5 seeds), change relative to the matched null *falls* as the update gets heavier, while change relative to the seed floor *rises* (Figure 2). For IG at shift magnitude 1.5 and update learning rate 2×10^{-4} , $\Delta/\rho_{\text{null}}$ runs 1.81, 1.58, 1.39, 1.28, 1.09 across 1, 2, 5, 20 and 60 epochs, while $\Delta/\rho_{\text{seed}}$ runs 0.11, 0.19, 0.41, 0.82, 1.07. That is the most

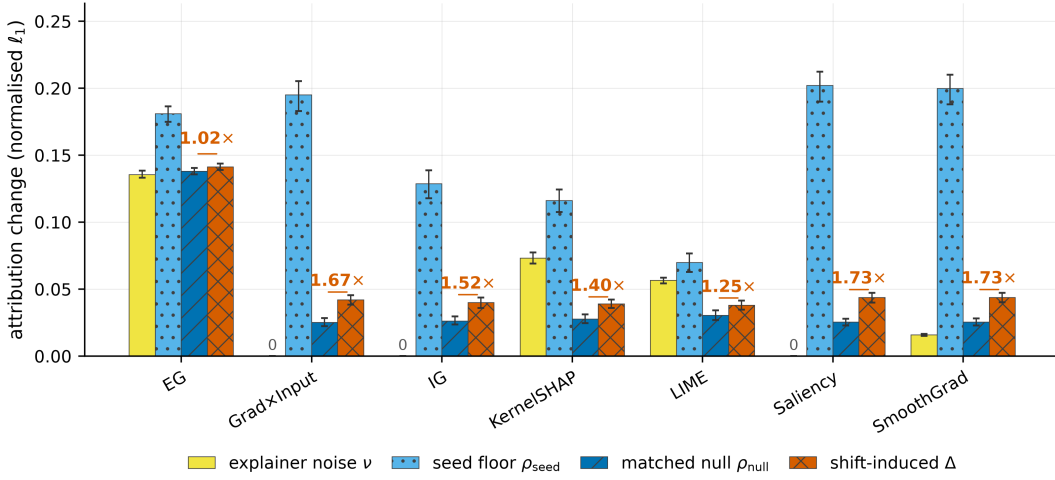


Figure 1: Attribution change against its floors under covariate shift, where ω is exactly zero, so any movement above the matched null is unwarranted. Bars are means over seeds with bootstrap intervals; brackets give $\Delta/\rho_{\text{null}}$. Deterministic explainers have $\nu = 0$ exactly, marked “0”. The seed floor ρ_{seed} exceeds the measured change for every explainer, which is the observation §6.2 turns on.

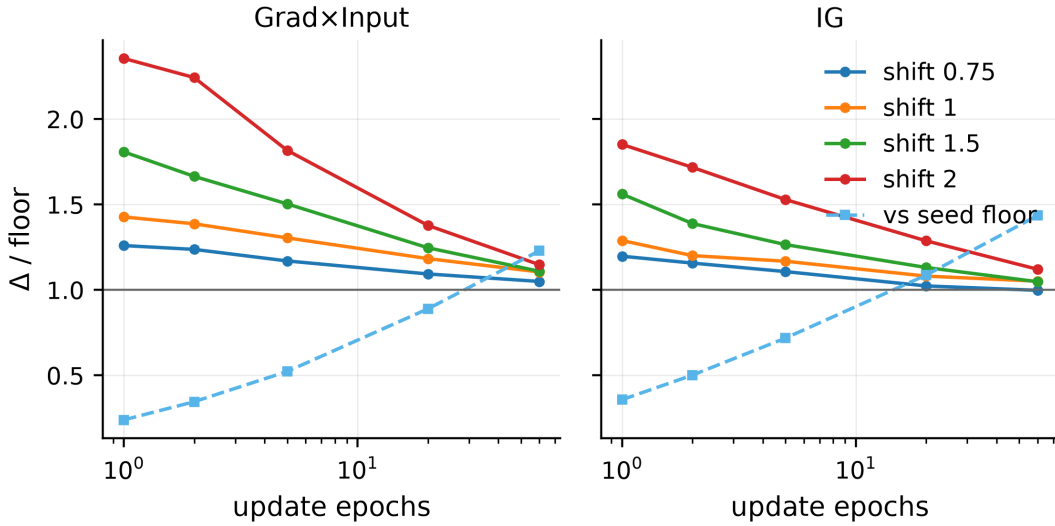


Figure 2: Change relative to the matched null (solid) and to the seed floor (dashed) as the update grows heavier. The two controls order the update regimes in opposite directions and the curves cross. Learning rates are pooled here; the single cell quoted in the text is more pronounced.

pronounced of the twelve cells, so we report what happens in the rest: the opposite ordering holds in 24 of 24 explainer $\times$ magnitude $\times$ learning-rate cells, and the curves cross in 10 of 12. The direction, which is what the argument needs, is unanimous. Read against the seed floor a heavy update looks maximally shift-driven; read against the matched null it is the least shift-driven condition. The reason is mechanical: the seed floor is a fixed quantity while Δ grows with training, so the ratio to it measures how hard one trained, not how far the data moved. Across 681 runs the matched null reproduces a median 82% of the movement that follows an update.

Stated plainly: against the seed floor our headline effect is inverted. In the covariate-shift condition of §6.3, $\Delta/\rho_{\text{seed}}$ runs 0.21 to 0.78 with a median of 0.31. A reader who takes seed retraining as the reference will conclude there is no phenomenon here. We put the number in the main text because it is the first thing a sceptical reader computes, and because our answer to it is the

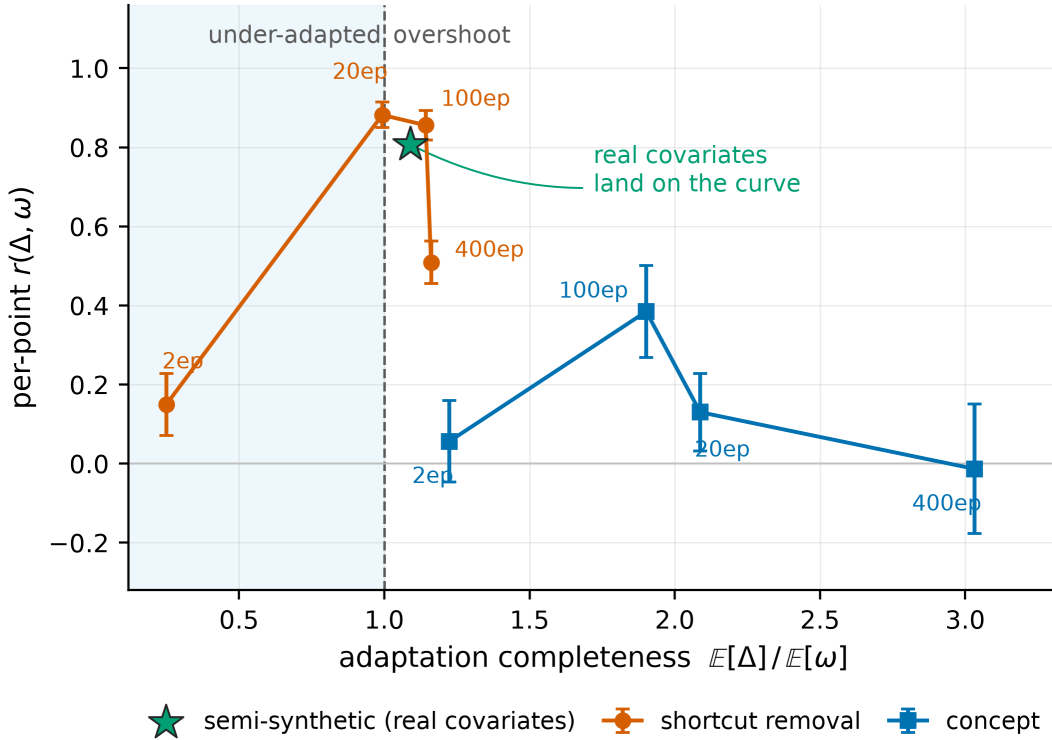


Figure 3: Per-point tracking of the warranted reference against how far the update actually travelled. Sweeping the update budget carries r from $+0.15$ to $+0.88$ as completeness passes through 1, then down again on overshoot. The semi-synthetic run on real covariates lands on the same curve, so the disagreement in §6.4 is not a property of the data.

paper’s argument and not a qualification of it. The obvious suspicion is that we adopted whichever control produced the answer we wanted. The placebo of §6.1 speaks against that, and it is decisive because its ground truth is fixed in advance: a control that manufactures a two-fold effect from an unshifted placebo cannot certify effects under shift. Two further points support it. The comparison is not like-for-like, since ρ_{seed} retrains from scratch on 8,000 points for 60 epochs while the update is 20 epochs on 4,000, so Δ/ρ_{seed} varies operator, data volume and step count at once. And the two controls disagree about direction, not only magnitude; a valid-but-conservative reference would preserve the ordering and change only the scale.

6.3 UNWARRANTED CHANGE EXISTS

Under covariate shift, where ω is exactly zero, all seven explainers move more than their matched null (Figure 1). The ratios are 1.73 (saliency and SmoothGrad), 1.67 (gradient $\times$ input), 1.52 (IG), 1.40 (KernelSHAP), 1.25 (LIME) and 1.02 (expected gradients), with Holm-adjusted $p = 0.014$ and Cliff’s δ up to $+0.98$. The effect is largest for gradient-boosted trees with exact TreeSHAP (2.08), so it is not a gradient artefact, and it persists on a small CIFAR ResNet. It also replicates on ACS Income, where 27 of 28 explainer $\times$ state cells remain significant after Holm correction. Ablating the distance, normaliser, ϵ threshold, IG baseline, SHAP background and probe sampling leaves the ordering of conclusions unchanged (Kendall $\tau = 1.00$ across the core axes).

6.4 MAGNITUDE AND LEGITIMACY: A CONDITIONAL RESULT

We now test whether the amount an explanation moves indicates whether it should have moved. Within a shift family each probe point carries its own ω , so measured change can be regressed on it directly. Correlations are computed within seed and aggregated across seeds. Under light updates the association is small: $r = +0.121$ [$+0.084, +0.157$] under concept shift over 3,237 points and

$r = -0.104 [-0.192, -0.015]$ under shortcut removal over 4,329 points, so the warranted change accounts for roughly 1% of the variance in the measured change. Both intervals exclude zero, so we do not claim the association is absent, only that it is small. We report this as a magnitude with an interval instead of testing against an equivalence margin, since any margin chosen after seeing these numbers would be the post-hoc move this paper criticises elsewhere.

The per-explainer version of the same comparison is underpowered at ten seeds. A two one-sided test (Schuirmann, 1987) against a ± 0.10 margin, fixed from the placebo’s own spread, declares only one of seven explainers equivalent, and that one is expected gradients, whose two ratios are both near 1.02 and whose pass therefore reports its insensitivity. LIME is a genuine exception: its ratio is 0.64 higher under shortcut removal than under covariate shift ($p = 0.002$). We name it instead of averaging it away, while noting that its own noise floor exceeds its matched null, so we cannot separate sensitivity from sampling variance.

The condition. Repeating the measurement on real ACS covariates with a mechanism known by construction gives $r = +0.81$, the opposite conclusion. The cause is not the covariates. Define completeness as $\mathbb{E}[\Delta]/\mathbb{E}[\omega]$, the fraction of the warranted distance the update actually covered; it is 0.12 and 0.54 in the two synthetic settings and 1.09 in the semi-synthetic one. Sweeping the update budget on the generator confirms the reading (Figure 3): on shortcut removal, r rises from +0.149 at completeness 0.249 to +0.882 at completeness 0.993, then falls to +0.509 once the update overshoots. Distance from perfect adaptation predicts tracking at $\text{corr}(|\log \text{completeness}|, r) = -0.870$, and on the rising limb completeness and r correlate at +0.756. The relationship is peaked and not monotone, which is why a naive pooled correlation of -0.356 would mislead. The claim is therefore conditional: attribution magnitude tracks warranted change when the update completes the adaptation, and under the light, prediction-preserving updates that deployment uses it carries little such information. That is the regime auditors work in, because heavy updates which complete the adaptation also move predictions visibly.

Two published metrics inherit the consequence. Reimplemented on the same checkpoints, a FASS-style filtered distance and Delta-Audit’s spurious residual rank correct adaptation as 45% and 89% worse than unwarranted drift, and UEC assigns them opposite signs.

6.5 RE-AUDITING A PUBLISHED RESULT

Applying the missing control to Hemmat & Fatemi (2025), we re-run its 45 reported settings with a matched null. 64% produce attribution movement smaller than refitting the same model on a resample of the same data, and only 33% clear that floor, with a median ratio of 0.861. Their own flagship examples survive comfortably, which is worth stating: the conclusion is about the reported settings in aggregate and not about the paper’s headline claims.

6.6 THE THEORY HOLDS, AND THE CHANGE IS NOT FAITHFULNESS LOSS

Proposition 1(i) records 0 violations in 20,000 probe points, once each run is compared against its own quadrature tolerance (Figure 4). Gradient $\times$ input breaks the same bound on 60.8% of points. The coalition premise of Proposition 3 fails off-manifold, with masked inputs moving 1.49 to 2.02 times further than data inputs across shift families. Separately, the effect is not explainers becoming wrong instead of merely different: restricting to points where the explainer is faithful to both checkpoints leaves the ratio unchanged (1.567 against 1.563), and the correlation between Δ and the faithfulness drop lies in $[-0.084, +0.032]$ across seed-averaged cells.

6.7 WHERE THE EFFECT STOPS

The effect is bounded, and we map the boundary and not the interior. It is flat at roughly 1.4 across a $630\times$ parameter range within models trained from scratch, moving from 1.48 at 1.7k parameters to 1.36 at 1.07M. On real census covariates with a mechanism known by construction it weakens to 1.20 [1.05, 1.34] for IG, and the machinery of ω transfers exactly there, matching quadrature to 1.1×10^{-14} on real rows under a tilt calibrated to the synthetic shift’s own domain-classifier AUC (0.906 against 0.902). On DistilBERT at 66.9M parameters it is absent: 1.02 [0.97, 1.09], unchanged across a tenfold range of update strength.

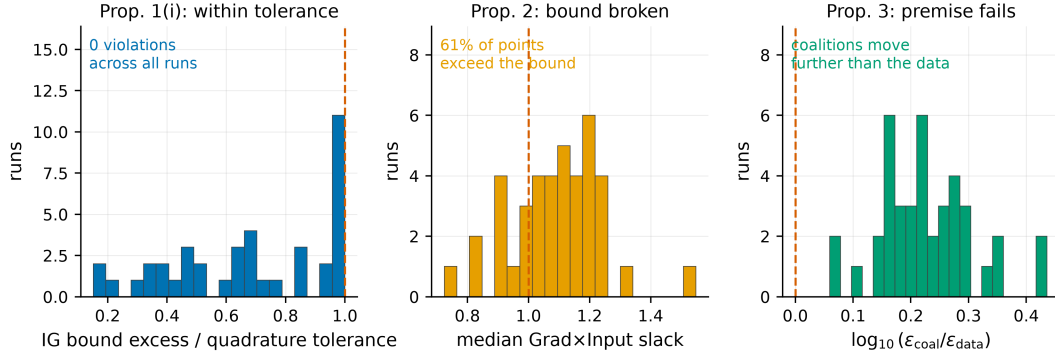


Figure 4: Per-run checks of the three propositions. Left: IG’s aggregate excess over the bound, divided by each run’s own quadrature tolerance, so values below 1 are numerical. Middle: gradient×input breaks the same bound. Right: the coalition premise of Proposition 3 fails off-manifold.

Modelling the optimiser share directly shows that the last result is not simple extrapolation (Figure 5, Appendix B). Capacity does not predict the share at all, with a logit-scale slope of -0.007 $[-0.197, +0.184]$, $p = 0.94$, across $630\times$ of parameters. Update strength does predict it, monotonically. At matched update size DistilBERT sits $+0.173$ $[+0.104, +0.243]$ above the from-scratch curve (Welch $p = 3.3 \times 10^{-4}$). Whether that excess comes from pretraining or from architecture, this design cannot separate.

Finally, the monitored quantity that does correlate with unwarranted change points the wrong way: higher prediction agreement between checkpoints goes with *more* unwarranted change, not less, and target accuracy’s association reverses sign across shift magnitudes.

7 LIMITATIONS

The reference needs a known Bayes predictor. ω is exact only where the generative process is known. On real data we claim the covariate-shift null alone, and even there it is checked through a calibration-transfer test and is not proved. The semi-synthetic construction removes the covariate half of that assumption but not the label half, which no method can remove.

H2 is conditional on the update completing the adaptation. We identify the governing quantity (§6.4) but characterise it on two shift families in one generator plus one semi-synthetic setting. Whether completeness governs tracking elsewhere is untested.

Scale. DistilBERT is covered, but nothing here speaks to vision transformers, and the $+0.173$ excess cannot be attributed to pretraining over architecture without a same-size transformer trained from scratch.

Our effect is smaller than the literature we critique. Against the seed floor it is inverted, and the case for the matched null rests on a placebo rather than on the shift results the control is used to interpret. A reader who rejects the placebo argument should reject the paper’s empirical claims with it.

8 CONCLUSION

The measurement of explanation change after a model update turns on a control that has not been used. Applying the identical update to fresh source data isolates the distribution from the optimisation, and it reverses the ordering that a seed-retraining floor produces. Most of the attribution movement reported after fine-tuning is reproduced by the optimiser alone. A reference for how far explanations should have moved makes the remaining question askable, and answers it conditionally: magnitude indicates legitimacy when an update completes the adaptation it implies, and carries little information under the light updates that deployment uses.

ETHICS STATEMENT

This work uses only public datasets and involves no human subjects, no personally identifying information and no new data collection. ACS Income (Ding et al., 2021) is derived from public census microdata and carries demographic attributes including sex and race, which our models consume as features because the published prediction task does. We make no fairness claim and use the data solely to test whether a measurement design transfers from a generator to real covariates; the attribution changes we report should not be read as evidence about any group. The paper is diagnostic in aim, and the plausible misuse we can foresee is that a practitioner reads unwarranted attribution change as a fairness finding, which §6.4 argues directly against. The authors have read the ICLR Code of Ethics and adhere to it.

REPRODUCIBILITY STATEMENT

Every quantitative claim is regenerated from the released code. A single entry point runs all fourteen stages in dependency order and rebuilds every table and figure. A verification script re-derives 80 headline numbers from the produced artefacts and fails if any drifts, and 101 tests cover the generator, the operators and a numerical check of each proposition. Figures write their source data beside the image. The synthetic and semi-synthetic pipelines run on CPU; the transformer arm requires one GPU and its measured output is committed so the rest reproduces without it. Code is supplied as anonymised supplementary material.

USE OF AI STATEMENT

Large language models were used as a coding and editing assistant: drafting and refactoring experiment code, proposing figure layouts, and copy-editing prose. All experimental design decisions, the choice of reference and controls, the propositions and their proofs, and every claim in this paper were made and checked by the authors. Every number reported here is produced by the released code and re-derived by the verification script described above.

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A TERMINOLOGY: WHAT “EXPLANATION DRIFT” DENOTES

The phrase names at least three distinct objects in work from 2023 onwards, which is why we avoid it. Mougan et al. (2023) use *explanation shift* for a divergence between $P(E_f(X_S))$ and $P(E_f(X_T))$ for a frozen model, a population-level object. Dhayalkar (2026) uses *explanation drift* for the epoch-to-epoch change of attributions on a fixed probe set within a single training run. The concept-drift literature uses *drift explanation* for the opposite direction of inference, explaining why the data moved (Hinder et al., 2023; Kulinski & Inouye, 2023). Our object is none of these: instance-level attribution change across a single discrete model update, measured on prediction-preserved probe points drawn from the shared support. We call it *explanation change* and reserve *drift* for temporal sequences, following the first use.

B THE OPTIMISER SHARE

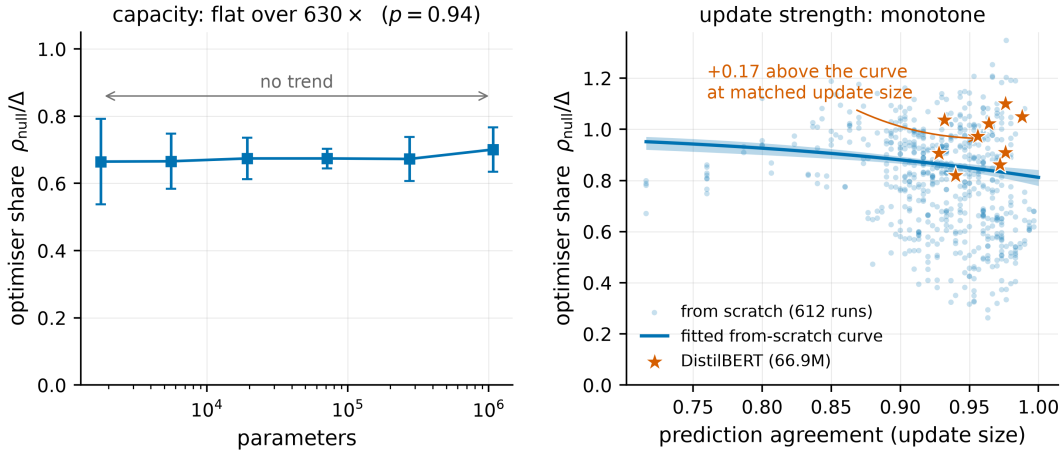


Figure 5: What predicts the share of movement the optimiser reproduces, $\rho_{\text{null}}/\Delta$. Left: capacity does not predict it, with a logit-scale slope of -0.007 $[-0.197, +0.184]$, $p = 0.94$, across a $630 \times$ parameter range. Right: update strength does, monotonically; the fitted from-scratch curve is shown with its 95% interval, and DistilBERT sits $+0.173$ $[+0.104, +0.243]$ above it at matched update size.

The share is $\rho_{\text{null}}/\Delta$: the fraction of measured attribution change that the matched null reproduces. A range on its own is a documented correlation; a model of it is a mechanism. Two covariates are comparable across model classes. Prediction agreement measures how far an update moved a model, and capacity measures its size. The width sweep holds the update fixed and varies parameters $630 \times$; the regime sweep holds capacity fixed and varies update strength. The share is bounded, so it is modelled on a logit scale; a linear fit on the raw share extrapolates to impossible negative values.

C FIGURE ACCESSIBILITY

Figures use the Okabe–Ito palette (Okabe & Ito, 2008) and encode every distinction redundantly, in hatch for bars and in marker shape for lines, so that they survive monochrome printing and the common colour-vision deficiencies. We simulated deuteranopia, protanopia and tritanopia by cone collapse and checked pairwise separation for the colours that share each panel; where separation on colour alone falls below threshold, the redundant encoding carries the distinction.